Sri Lanka Institute of Information Technology

Faculty of Computing — Department of Computer Systems Engineering

IE3122 – Network Security

Week 4 Tutorial Worksheet: Layer 3 Attacks & Defense

Field	Detail
Based on	Week 4 — Layer 3 Attacks & Defense lecture (ICMP overview, ICMP-based attacks, IP-based attacks, mitigations)
Learning Outcomes	LO1 (describe network vulnerabilities/attacks) and LO2 (evaluate mitigation techniques)
Format	10 structured questions, no MCQs — answer in your own words
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic — use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's Layer 3 lecture. Several ask you to compare or evaluate rather than just describe — that's deliberate: LO2 is specifically about weighing mitigation options, not just naming them. Write full answers, not one-line definitions.

ICMP Fundamentals

Q1. Explain why ICMP has no authentication, and why that design choice — while necessary for normal network diagnostics like ping and traceroute — creates the vulnerability every attack this week exploits. *[LO1]*

ICMP-Based Attacks

Q2. Describe how ICMP Sweep works, and explain why it's typically the first step in a larger attack rather than a goal in itself. *[LO1]*

Q3. Distinguish Ping Flood from the Smurf attack — both are ICMP-based denial-of-service attacks, but what's the key mechanism difference, and why is Smurf considered more dangerous per unit of attacker effort? *[LO1]*

Q4. A network administrator is deciding whether to block all inbound ICMP at the perimeter entirely, or rate-limit it instead. Evaluate this trade-off, referencing at least two of today's attacks in your answer. *[LO2]*

IP-Based Attacks

Q5. Explain how IP Spoofing works, and why ingress/egress filtering (BCP 38) is effective against it. *[LO1, LO2]*

Q6. Distinguish the TTL Expiry Attack from the TTL-Based Evasion Attack — both abuse the TTL field, but what does each actually try to achieve, and who/what is the real target in each case? *[LO1]*

Q7. Describe how the Teardrop attack exploits IP fragmentation, and explain why this is considered a largely legacy attack against modern systems today. *[LO1]*

Bringing It Together

Q8. IP Spoofing and the Smurf attack share a common root cause. Identify it, and explain why a single mitigation (ingress/egress filtering) is effective against both. *[LO1, LO2]*

Q9. Compare Ping of Death and Teardrop — what do they have in common in terms of the type of vulnerability they exploit and the mitigation strategy that addresses them? *[LO2]*

Q10. Of all the attacks covered today, choose the one you think would be hardest for a traditional signature-based IDS to detect, and justify your choice. *[LO2, LO3]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. Several of today's mitigations (ingress/egress filtering, patching, rate limiting) repeat across multiple attacks — notice which ones do double duty, since that's exactly the control-minimisation thinking from Week 2's layered security plan.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1–LO2) and final exam (MCQ + structured essay, LO1–LO5) — the evaluation-style questions above are close in format to the structured-essay section.